

Quantum Mechanics II Midterm Exam 1

Question 1) Starting from the time-dependent Schrödinger equation in operator form,

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle,$$

where $\hat{H}$ is the Hamiltonian operator, derive the time-dependent Schrödinger equation in the energy eigenbasis assuming a discrete energy spectrum. Do the same for an infinite dimensional Hilbert space with a continuous energy spectrum.

Question 2) a) The propagator is defined as the amplitude for a particle to propagate from position x_1 at time t_1 to position x_2 at time t_2 :

$$K(x_2, t_2; x_1, t_1) = \langle x_2 | \hat{U}(t_2, t_1) | x_1 \rangle,$$

where $\hat{U}(t_2, t_1)$ is the time evolution operator. Derive the composition property of the propagator:

$$K(x_3, t_3; x_1, t_1) = \int dx_2 K(x_3, t_3; x_2, t_2) K(x_2, t_2; x_1, t_1).$$

b) For a free particle in one dimension with mass $m = 1$ and initial position and time $x_1 = 0$ and $t_1 = 0$, the propagator can be calculated explicitly and is given by

$$K(x, t; 0, 0) = \sqrt{\frac{1}{2\pi i t}} \exp\left(\frac{ix^2}{2t}\right).$$

Use this expression to verify the propagator satisfies Schrödinger's equation for a free particle with respect to both x and t .

Question 3) a) Prove the Feynman-Hellmann theorem: if $\hat{H}(\lambda)$ is a Hamiltonian that depends on a parameter λ , and $|n(\lambda)\rangle$ is an eigenstate of $\hat{H}(\lambda)$ with eigenvalue $E_n(\lambda)$, then

$$\frac{dE_n(\lambda)}{d\lambda} = \langle n(\lambda) | \frac{d\hat{H}(\lambda)}{d\lambda} | n(\lambda) \rangle. \quad (1)$$

You can assume that the eigenstates $|n(\lambda)\rangle$ are normalized and that the eigenvalues $E_n(\lambda)$ are non-degenerate.

b) Now use this result to express the first order correction to the energy levels of a system under a perturbation $\lambda\hat{V}$ by Taylor expanding the energy levels $E_n(\lambda)$ around $\lambda = 0$ and expressing your answer in terms of the unperturbed eigenstates $|n(0)\rangle$ and the perturbation $\hat{V}$.

MORE QUESTIONS ON THE NEXT PAGE

**** **Question 4)**¹ Imagine that the photon had a mass m_γ . The Coulomb potential would then be modified to a Yukawa potential:

$$V(r) = -\alpha \frac{e^{-m_\gamma r}}{r}.$$

a) For very small m_γ , what is your perturbation to the Hydrogen atom Hamiltonian? Calculate the *first* and *second* order corrections to the energy levels of the hydrogen atom by expanding the Yukawa potential in powers of $m_\gamma r$.

Hint: you can use

$$\langle 1 \rangle = 1, \quad \langle r \rangle = \frac{a_0}{2} [3n^2 - l(l+1)]$$

b) Under what conditions is the perturbation theory valid? Express your answer in terms of an inequality for m_γ , α , and m_e or a_0 . (You only need to consider the first order correction for this part.)

c) Why can we not use the $\mathcal{O}(m_\gamma)$ correction to set a bound on m_γ by measuring transition energies?

d) Including your correction of $\mathcal{O}(m_\gamma^2)$, provide an upper bound on the mass of the photon in eV given that the energy difference between $1s$ and $2s$ states of hydrogen is measured to be $\Delta E \simeq 10.2$ eV with a precision of one part in 10^{12} .

e) Provide a qualitative discussion of potential obstacles that would prevent us from using this method to set a bound on the photon mass with such high precision. What are some other effects that we may be missing once such precision is considered?

(Hint: $\alpha = 1/137 \simeq 10^{-2}$, $a_0 = \frac{1}{m_e \alpha}$, $m_e \simeq 0.5$ MeV.)

Question 5) a) In the interaction picture, show that the time evolution operator can be expressed as

$$\hat{U}_I(t, t_0) = e^{i\hat{H}_0 t} U(t, t_0) e^{-i\hat{H}_0 t_0}.$$

where $U(t, t_0)$ is the time evolution operator in the Schrödinger picture and $\hat{H}_0$ is the free Hamiltonian.

b) Starting from the ansatz for a wavefunction in the interaction picture,

$$|\psi_I(t)\rangle = \sum_n c_n(t) |n\rangle,$$

where $|n\rangle$ are eigenstates of the free Hamiltonian $\hat{H}_0$, show that the coefficients $c_n(t)$ satisfy the following equation:

$$i\hbar \frac{dc_n(t)}{dt} = \sum_m \langle n | \hat{V}_I(t) | m \rangle c_m(t),$$

where $\hat{V}_I(t) = e^{i\hat{H}_0 t} \hat{V} e^{-i\hat{H}_0 t}$ is the interaction picture potential. Without actually deriving it, describe how you would use perturbation theory to solve this equation for $c_n(t)$.

¹This question is mandatory. I want to see you plug in numbers.